Stormwater Regulation Summary: Washington, DC

Municipal Stormwater Management Technical Profile | Sempergreen USA

1. Regulatory Overview

- Primary driver: Chesapeake Bay TMDL compliance and local water quality improvement.
- DC Stormwater Management Rule (2020 update) administered by DOEE (Dept. of Energy & Environment).
- Applies to projects with ≥5,000 SF of land disturbance.
- Unique feature: Tradable Stormwater Retention Credits (SRCs) create a market-based incentive for on-site retention.
- Green Area Ratio (GAR) requirements in most zoning districts push vegetated solutions including green roofs.

2. Typical Soil Conditions

- Piedmont/Coastal Plain transition zone — soils range from clay-rich residual soils in upper NW to alluvial deposits along the Anacostia and Potomac.
- Hydrologic Soil Group: Predominantly C and D in developed areas; some B soils in less-disturbed upland areas.
- Heavy Piedmont clays (Christiana series and similar residual soils) limit infiltration in many parts of the city.
- Implication: Infiltration-based BMPs face challenging soil conditions — retention and controlled release (including rooftop) are preferred.
- *Site-specific soil investigation required; alluvial areas near waterways may have different characteristics.*

3. Green Roof Requirements

- Green roofs are not strictly mandatory, but GAR (Green Area Ratio) requirements in most zones create strong incentive.
- GAR assigns landscape multipliers — green roofs receive a multiplier of 0.60 (extensive) to 0.80 (intensive), making them efficient for compliance.
- DOEE RiverSmart Rooftops program provides rebates for green roof installation.
- Green roofs generate SRCs that can be sold on the open market (average ~$1.40/gal/yr as of 2024; prices fluctuate) — creating ongoing revenue.
- Clean Energy DC targets 100% renewable by 2032; bio-solar (green roof + PV) addresses both stormwater and energy goals.

4. TSS Requirements

- DC stormwater rule requires 60% TSS removal for general runoff; 80% TSS removal required for vehicular surfaces (parking, driveways) draining to MS4 or waterways.
- Green roofs are an approved BMP credited with TSS removal — meeting the 60% general standard for captured volume.
- Phosphorus and nitrogen removal also required under Chesapeake Bay TMDL — green roofs receive pollutant removal credits for both.
- SRC calculations include pollutant removal benefits, increasing the credit value of green roofs vs. detention-only systems.
- *TSS removal credits may vary by green roof design — verify against current DOEE Stormwater Management Guidebook.*

5. Nature-Based Retention Requirements

- New development: Retain 1.2" SWRv (Stormwater Retention Volume) from all impervious surfaces.
- Major renovation: Retain 0.8" SWRv.
- SWRv must be achieved through on-site BMPs — infiltration, evapotranspiration (green roofs), or rainwater harvesting.
- Alternatively, retention credits can be purchased via SRC market or in-lieu fee ($4.71/gal as of Aug 2024, adjusted annually for CPI; applies to first 50% of requirement).
- CN (Curve Number) method used for volume calculations; typical urban post-development CN: 98 (impervious) reduced to 55-70 with BMPs.
- On-site retention is almost always more cost-effective than SRC purchase over the building lifecycle.

6. Allowable Outflow Rates

- 2-year storm: Post-development peak must not exceed pre-development peak discharge.
- 15-year storm: Same peak flow control requirement.
- DC does not specify a flat l/s/ha rate — allowable discharge is site-specific based on pre-development conditions.
- *Typical pre-development release rates for urban sites range from 30-80 l/s/ha depending on existing imperviousness.*
- Extended detention required for channel protection: 24-hour drawdown of the 1-year storm event.